

KRISTU JAYANTI COLLEGE (AUTONOMOUS) BENGALURU
PG END SEMESTER THEORY EXAMINATION APRIL/MAY 2025 [1PAPR25]
MCA/MSC [CS] IV SEMESTER
MCCA04A41:QUANTUM COMPUTING

TIME: 3 HOURS

MAX. MARKS: 70

SECTION – A (Answer any five questions. Each answer carries four marks.) **[5 x 4 = 20]**

Q.No.	Questions	CO	BL
1.	A. What is Ket notation in quantum computing? B. Give the representation of the state $ 001\rangle$. (2+2)	CO1	L1
2.	Given the qubit state: $ \psi\rangle = \frac{1+i}{2} 0\rangle + \frac{1-i}{2} 1\rangle$ Is this a valid qubit state? Justify your answer.	CO1	L2
3.	A. Write the matrix representation of the Pauli-X gate. B. Determine the output quantum state when the Pauli-X gate is applied to: i) $ 0\rangle$ ii) $ 1\rangle$ (1+3)	CO2	L2
4.	A. What is the Hadamard gate used for in quantum computing? B. Write the matrix form of the Hadamard gate and show its action on the states $ 0\rangle$ and $ 1\rangle$. (1+3)	CO2	L2
5.	Write a short note on key distribution protocol.	CO3	L1
6.	List any four primary sources of error in quantum computing.	CO4	L2
7.	What are error sub spaces in Quantum Error Correction system?	CO4	L2
8.	What are some positive societal impacts of Quantum Computing?	CO5	L2

SECTION – B (Answer any five questions. Each answer carries ten marks.) **[5 x 10 = 50]**

Q.No.	Questions	CO	BL
9.	A. What is quantum entanglement? B. what are Bell states in this context? C. how can all four Bell states be mathematically derived and proven. (2+2+6)	CO1	L3
10.	A. What is the principle of quantum superposition? B. Determine whether the following quantum states are valid superposition states. Justify your answers with proper calculations. $ \psi\rangle = \frac{3}{5} 0\rangle + \frac{4}{5} 10\rangle$ $ \phi\rangle = \frac{2}{3} 0\rangle + \frac{1}{3} 0\rangle$ (4+6)	CO1	L4
11.	A. Describe the functionality of the following commonly used two-qubit gates with matrix representation: i. Controlled-NOT (CNOT) gate ii. SWAP gate B. Apply a CNOT gate to the input state $ \psi\rangle= 10\rangle$ Show the resulting state. (7+3)	CO2	L4

12.	Obtain the output of given quantum circuit.	CO2	L4
13.	Explain the concept of quantum key distribution (QKD) and its role in quantum secure communication.	CO3	L3
14.	Describe the Shor code and explain how it is used for quantum error correction.	CO4	L2
15.	Explain the architecture of Quantum error correction system.	CO4	L2
16.	Discuss Privacy, security, and inequality in era of Quantum Computing.	CO5	L3

CO - COURSE OUTCOMES

BL - BLOOM'S TAXONOMY LEVEL

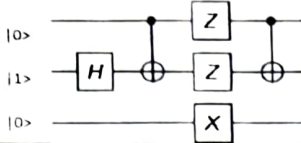
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